3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one "shall" sentence	Priority	Source	Acceptance criterion
FR-01	The system shall allow a new user to create an account using an email address and a password of at least 8 characters	MUST	Students (2.2), Scope (1.2)	A tester registers with a valid email and an 8 characters password and can then log in. A 7 character password is rejected with an error message.
FR-02	The system shall reject a login attempt whose email and password do not match a registered account.	MUST	Students (2.2); Scope (1.2)	A tester enters a registered email with a wrong password. Access is denied and an error message appears. Correct credentials grant access.
FR-03	The system shall allow a logged-in student to create an application record containing company name, position title, and date applied.	MUST	Students (2.2); Scope (1.2)	A tester creates a record with all three fields. It appears in the application list with the values entered. Submitting with the company name blank is rejected.
FR-04	The system shall allow a student to edit any field of an existing application record.	MUST	Scope (1.2)	A tester changes the position title of a saved record. The new value shows after a page reload.
FR-05	The system shall delete an application record only after the student confirms the deletion in a confirmation prompt.	MUST	Scope (1.2)	Choosing "Cancel" leaves the record in the list. Choosing "Confirm" removes it, and it stays removed after a reload.
FR-06	The system shall restrict an application's status to exactly one of Applied, Interviewing, Offered, or Rejected.	MUST	Scope (1.2); Definitions (1.3)	The status control offers only these four values. A new record defaults to Applied.
FR-07	The system shall record the previous status, new status, and date of every status change on an application.	MUST	Scope (1.2)	A tester changes a record from Applied to Interviewing to Offered. The history view lists two entries in chronological order with correct dates.
FR-08	The system shall allow a student to store an application deadline and one or more interview dates on an application record.	MUST	Students (2.2); Scope (1.2)	A tester saves a deadline and two interview dates. All three show on the record after a reload.
FR-09	The system shall display a reminder for any application deadline or interview date in the next 7 days	SHOULD	Students (2.2); Scope (1.2); C3 (2.4)	A tester saves 2 applications with deadlines 3 and 10 days away respectively. Only the 3 day deadline appears in the reminders
FR-10	The system shall display a dashboard summary of the total applications and the count of applications in each status (Applied, Interviewing, Offered, Rejected)	MUST	Students (2.2); University Advisors (2.2); Scope (1.2); Definitions (1.3)	A tester creates 4 applications, one of each status. The dashboard shows the total count as 4 and 1 for each status

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-11	The system shall allow a student to search applications by company name or position title and filter by application status.	SHOULD	Students (2.2); Scope (1.2)	Given 5 saved applications across different companies and statuses, by filtering using “Interviewing” or by searching by company name it displays only the matching records
FR-12	The system shall allow a student to upload and attach a PDF resume and cover letter of up to 10MB each to an application record	COULD	Students (2.2); Scope (1.2)	A tester uploads a 2 MB PDF resume. The file stays attached and downloadable. A 15MB file or a filetype that isn’t PDF is rejected.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	95% of dashboard page loads shall complete within 2 seconds for an account holding up to 200 application records, measured over 100 loads on a standard broadband connection.	SHOULD	Students (2.2)	Seed a test account with 200 records. Time 100 loads with browser developer tools and compute the 95th percentile.
QR-02	100% of requests by one user to view, edit, or delete another user’s application record shall be denied.	MUST	Students (2.2)	Run a scripted set of at least 20 cross-account requests (direct URLs and API calls) from a second test account. Every request must return an access-denied response.
QR-03	No user password shall be stored in plaintext. Passwords shall be stored only as salted hashes produced by bcrypt or Argon2.	MUST	Students (2.2)	Register 5 test accounts, inspect the user table, and confirm that no stored value matches any entered password and every value has the hash algorithm’s format.
QR-04	At least 4 of 5 first-time users shall create their first application record within 3 minutes without help.	SHOULD	User characteristics (2.3)	Run a moderated usability session with 5 students who have not seen the system, and time each from login to saved record.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall authenticate users only through application-managed accounts and shall not use university single sign-on.	C1 (2.4); Scope (1.2)	University SSO needs approval from the university’s IT office, which the team cannot get within the course timeline.
CON-02	The system shall not connect to any employer portal or job board to submit applications or retrieve their statuses.	C2 (2.4); Scope (1.2)	These sites need credentials or APIs the team has no access to, and automated access may violate their terms of service.

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-03	The system shall display deadline and interview reminders only in the web interface and not send text or email notifications	C3 (2.4); Scope (1.2)	External notifying requires third party messaging services and paid APIs that are not within the scope's boundaries.
CON-04	The system shall be delivered only as a web application and not built as a mobile app	C4 (2.4); Purpose (1.1)	Building and maintaining a mobile app alongside a web app exceeds the team's schedule constraints for the semester

3.4 Initial use cases

